Rachit Shrivastava

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Summary

Computer Science undergraduate (class of 2027) with a strong interest in backend and distributed systems. Experienced in building event-driven applications with Java, Spring Boot and Apache Kafka, and in taking full-stack projects from code to deployment using React, Docker and cloud hosting. Solid grounding in data structures, operating systems and databases, with added exposure to machine learning through a multi-agent reinforcement learning capstone.

Education

Vellore Institute of Technology — Bhopal, Madhya Pradesh
B.Tech in Computer Science — Graduation: 2027

Smart Traffic Management (Capstone): co-developed a traffic-signal control study using multi-agent RL in the SUMO simulator (Python), with automated data generation.
Evidence Preservation & AI-Assisted Triage for a Cybercrime Console (EPICS Group 608, co-author): contributed to a standards-based architecture paper covering hashing, WORM storage, append-only chain-of-custody logs and zero-trust access.

Technical Skills

Programming Languages: Python, Java
Frameworks & Technologies: Spring Boot, Apache Kafka, React, AWS
AI Tooling & Integration: Model APIs (OpenRouter, Hugging Face Hub), AI-Assisted Prototyping & Development Tools
Core Computer Science: Data Structures & Algorithms, Operating Systems (CPU Scheduling, Memory Management), Database Management Systems (Relational Database Design), Computer Networks
Engineering Practices: Object-Oriented Programming (Java), System Design Fundamentals, Software Engineering
Developer Tools: Git, Docker, VS Code

Projects

Midas Core – Transaction System | *Spring Boot, Kafka, React* — January 2026
Live Demo: https://midas-frontend-dl4i.onrender.com

- Engineered an event-driven transaction engine using Apache Kafka and the Transactional Outbox pattern to ensure exactly-once processing and zero message loss.
- Implemented real-time WebSocket architecture to instantly push transaction status updates to cross-client React interfaces.
- Secured full-stack application with Google OAuth 2.0 and JWTs, and deployed Dockerized microservices to Render.

Amazon Scout – Price Tracking Extension | *JavaScript, Node.js, Express, REST APIs*

- Built a Chrome/Edge extension that tracks product prices, backed by a Node.js and Express API.
- Integrated a third-party search API to compare prices across sellers and show them in a dashboard.

Operating System Simulator | *Python, Algorithm Design* — 2024 – In Progress

- Building an educational Python-based project to understand OS concepts through hands-on implementation of scheduling algorithms.
- Developing simplified simulations of CPU scheduling (FCFS, Round Robin) and basic virtual memory management routines.

Certifications

- Marketing Analytics, NPTEL (IIT Kharagpur), Elite, 90% — Jan–Apr 2026
- Introduction to Machine Learning, NPTEL — 2025
- Google IT Support Certificate, Google Career Certificates — Mar 2026
- The Bits and Bytes of Computer Networking, Google (Coursera) — Jan 2026